American Lobster (*Homarus americanus*)

Summary Scores

- **Sensitivity:** High - Very High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Gulf of Maine/Georges Bank Score	Gulf of Maine/Georges Bank LH DQ	Gulf of Maine/Georges Bank Tally	Southern New England Score	Southern New England LH DQ	Southern New England Tally
Adult Mobility	2.44	2.86		2.4	2.89		2.48	2.83	
Complexity in Reproductive Strategy	2.4	2.52		2.28	2.52		2.52	2.52	
Dispersal of Early Life Stages	1.68	2.75		1.64	2.9		1.72	2.6	
Egg and Larval Survival and Settlement Requirements	2.16	2.9		2.12	2.9		2.2	2.9	
Historical Variability in Temperature	2.54	2.85		2.44	2.86		2.64	2.84	
Juvenile Habitat Requirements	1.78	2.87		1.68	2.94		1.88	2.8	
Other Stressors	3.4	2.72		3.16	2.79		3.64	2.65	
Population Growth Rate	3.64	2.71		3.64	2.75		3.64	2.68	
Prey Specificity	1.12	3		1.12	3		1.12	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	3.5	2.9		3.52	2.9		3.48	2.9	
Spawning Cycle	2.8	2.65		2.76	2.7		2.84	2.6	
Stock Size/Status	3.02	1.84		2.48	1.84		3.7	1.84	
Total Sensitivity | Certainty	**High (3) | 0.83**			**High (3) | 0.99**			**Very High (4) | 0.92**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.43	1.13	
Bottom Oxygen	2.83	0.98	
Bottom Salinity	2.68	1.01	
Bottom Aragonite Solubility	1.31	0.76	
Net primary production	1.05	0.16	
Downward Flux of Particulate Organic Carbon	1.33	0.34	
Total Exposure	**Moderate (2.11)**	**0.95**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (6.33)**	**2.85**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for American Lobster. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate	High - Very High	Increase
Exposure	High	Moderate	Decrease
Vulnerability	Moderate	Moderate	None
Directional Effect	Neutral	Negative	Decrease

Stocks

American Lobster is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

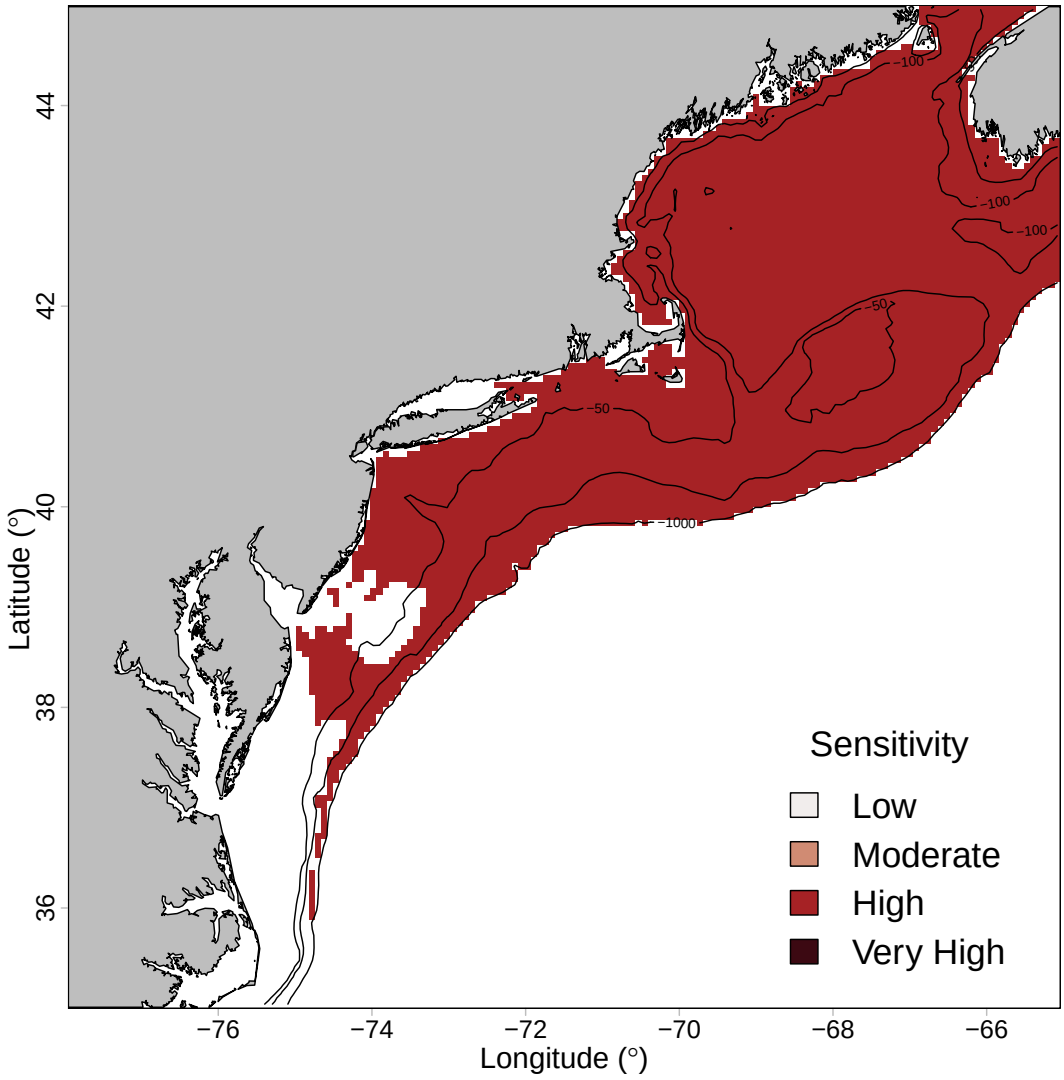


Figure 2: American Lobster Sensitivity on the Northeast Shelf

The **Gulf of Maine/Georges Bank** stock of American lobster received a **High** sensitivity score and the **Southern New England** stock scored **Very High.** Certainty was high for both stocks, with 99% certainty for the Gulf of Maine/Georges Bank stock and 92% certainty for the Southern New England stock (Figure 1).

Population growth rate and other stressors drove the scores for both stocks. Relationships to calcified organisms contributed to the High score for the Gulf of Maine/Georges Bank stock, and received a score of 3.5 for the Southern New England stock. Stock size had the highest score for the Southern New England Stock (Figure 1).

Lobsters are long-lived, slow growing species that can reach ages of 50+ years. Disease and the introduction of new predators such as black sea bass and spiny dogfish impact lobsters across their range (Atlantic States Marine Fisheries Commission (ASMFC) 2025; Goldstein et al. 2025). American lobsters are omnivores but diets are dominated by mollusks (Elner and Campbell 1987), which are likely to be impacted by ocean acidification.

Sensitivity has increased since the last assessment, where lobsters received a Moderate score. Population growth rate and other stressors drove scores in both assessments (Hare et al. 2016).

Exposure:

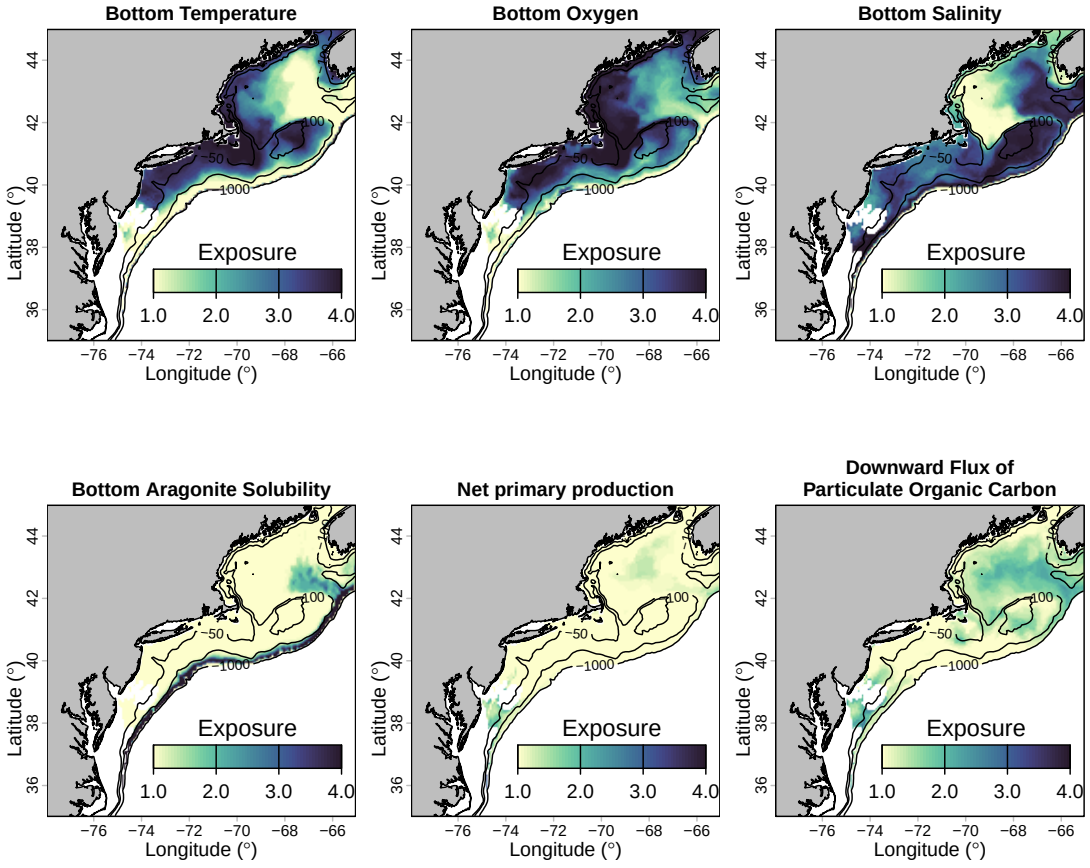


Figure 3: Maps of each Exposure Factor for American Lobster. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for American lobster across its range was **Moderate** (2.11 +/- 0.95) (Figure 1). Exposure was Very High over Georges Bank and High in Southern New England and along the coast of the Gulf of

Maine. Exposure was also High along the continental shelf break (Figure 4). These patterns were driven by bottom temperature, oxygen, salinity, and aragonite saturation exposure (Figure 3).

In the previous Assessment, American lobster received a High exposure score driven by sea surface temperature and ocean acidification (Hare et al. 2016).

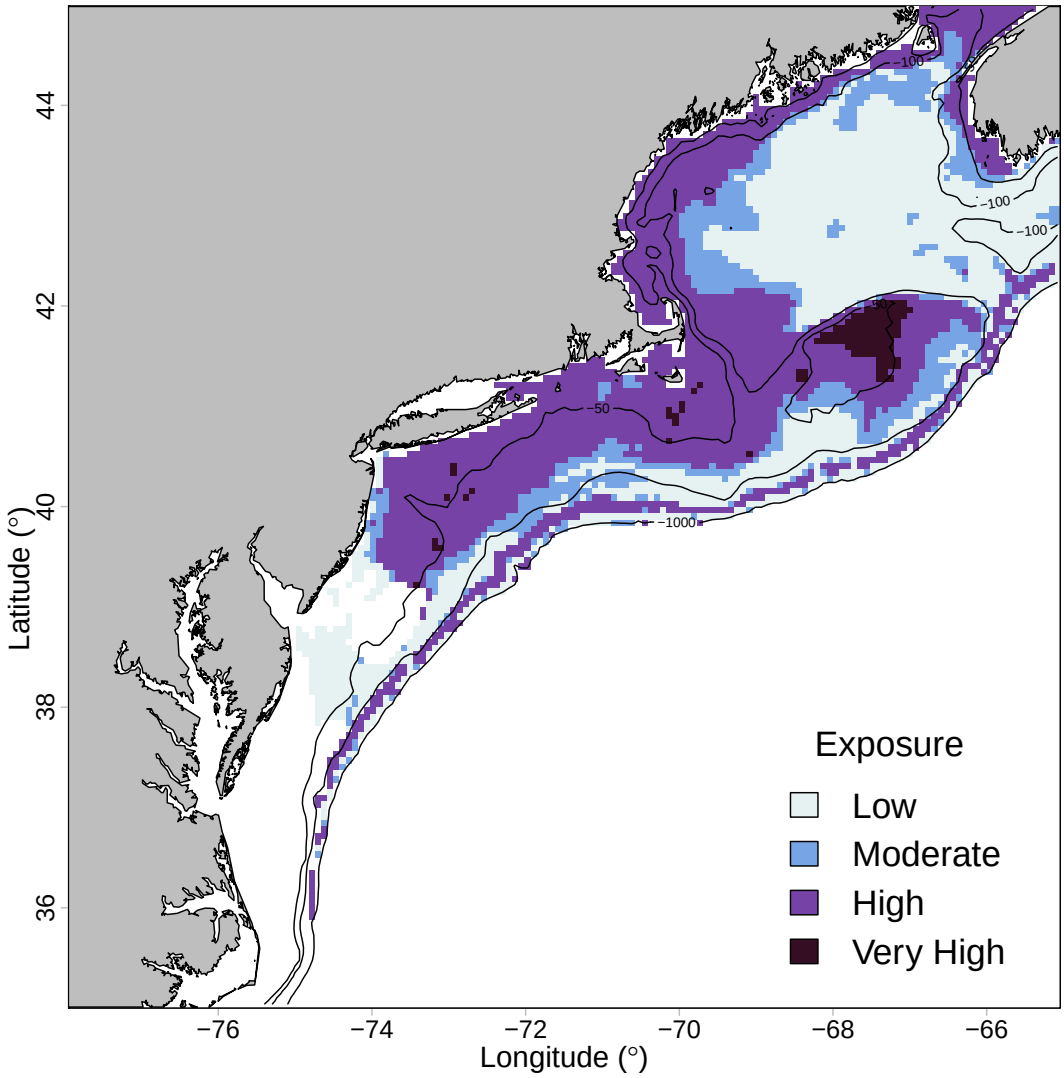


Figure 4: Total Exposure on the Northeast Shelf for American Lobster. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

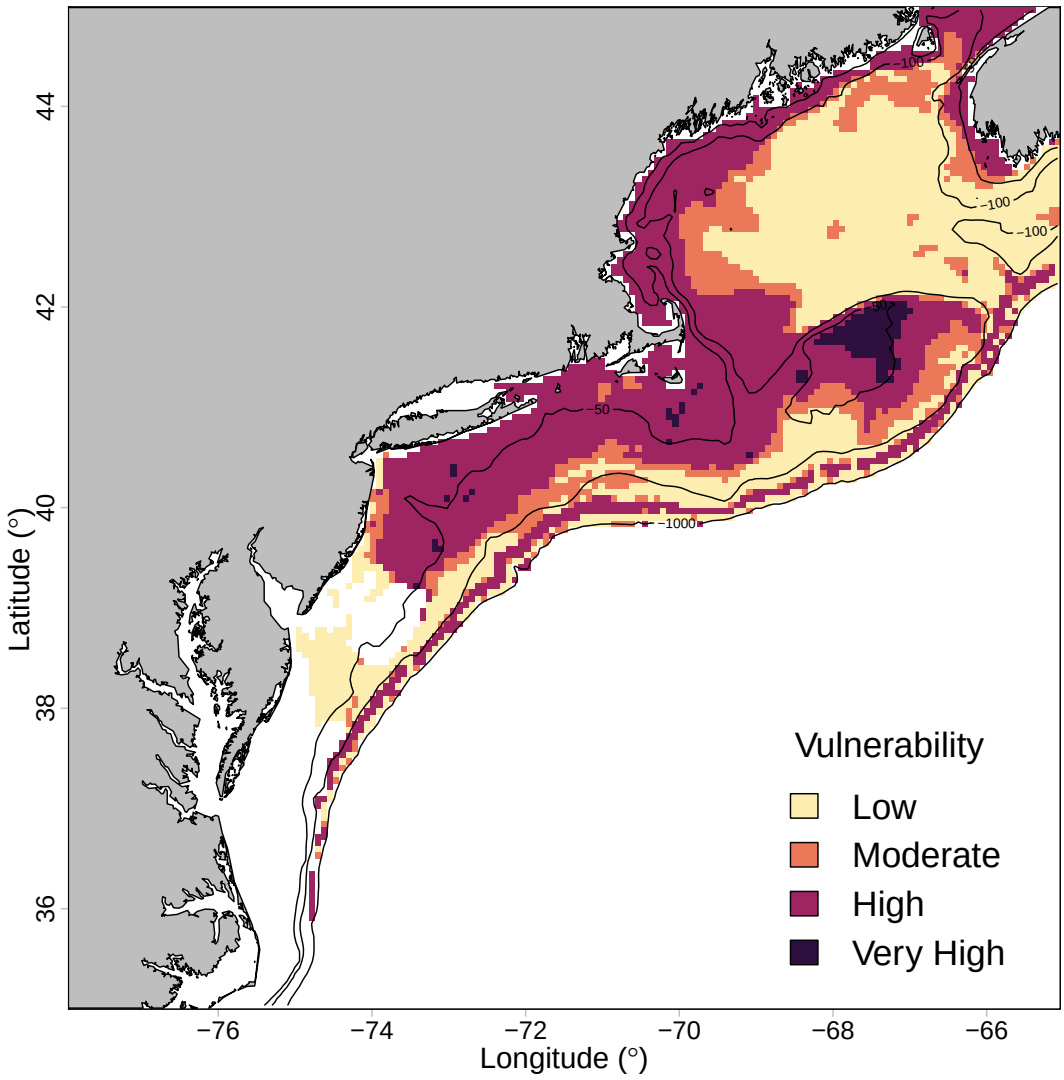


Figure 5: Vulnerability on the Northeast Shelf for American Lobster. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean vulnerability for the American lobster was **Moderate** (6.33 +/- 2.85) across its range (Figure 1). Vulnerability was Very High over Georges Bank and High along the coast of the Gulf of Maine, in Southern New England, and along the continental shelf break (Figure 5).

In the previous Assessment, American lobster also received a Moderate Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

The percent of data quality scores greater than 2 was 97% for the Gulf of Maine/Georges Bank stock and 95% for the Southern New England stock, indicating excellent data quality. For both stocks, Stock Size/Status was the only data gap (mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

The mean model confidence score for American Lobster was 1.45 (+/- 0.45), indicating that experts had low confidence in the model. Experts stated that the model missed seasonal inshore and offshore migrations. The ensemble model's AUC was 0.86, indicating good predictive performance.

Additional Metrics

Distribution Shifts:

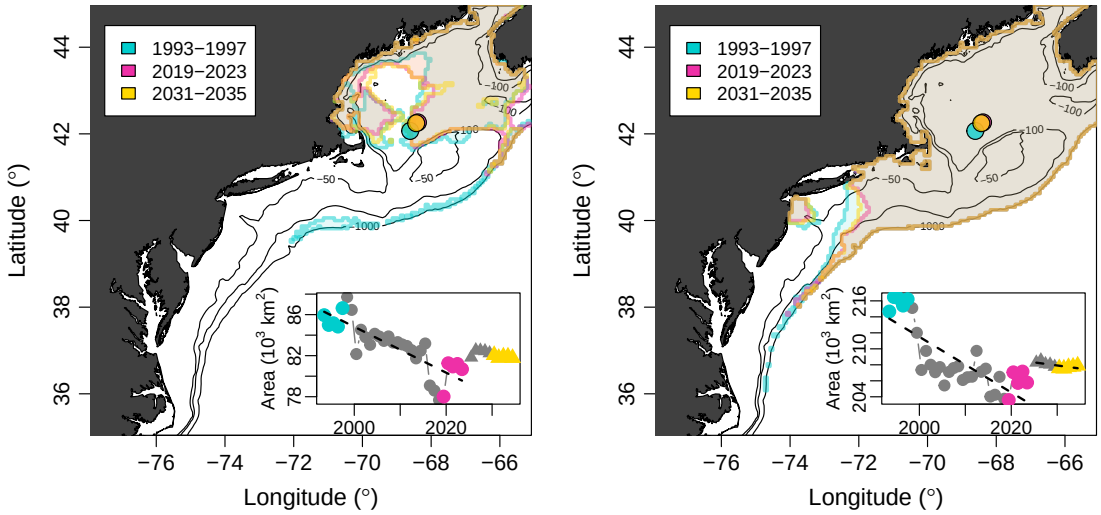


Figure 6: Distribution shifts on the Northeast Shelf for American Lobster. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Between 1993 and 2023, the weighted centroid shifted nearly 29 km northeast. There were significant declines in lobster habitat along the southern and western edges of their distribution (Figure 6).

The American lobster weighted centroid is predicted to shift 4.5 km west-southwest under near future ocean conditions. Distributions are expected to expand westward and south (Figure 6).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	85,851	215,882	4.74	7.30
2019-2023	28.6 km NE	80,403 (-5,447)	205,927 (-9,955)	4.36 (-0.37)	6.73 (-0.56)
2031-2035	4.5 km WSW	81,912 (+1,509)	207,712 (+1,784)	4.36 (0)	6.81 (+0.08)

Directional Effect:

Both stocks are likely to be **negatively** impacted by climate change (certainties range from 81 – 100% based on bootstrap analysis of expert scores). Increased thermal stress and disease will continue to decrease the productivity of the stocks. In addition, new demersal predators such as black sea bass and spiny dogfish will increase predation pressures of juvenile and adult lobster. The directional effect was estimated to be neutral in the previous Assessment (Hare et al. 2016).

Climate Effects on Abundance and Distribution

The Northwest Atlantic has been warming at an exceptionally fast rate (Saba et al. 2016), which has contributed to declines in fecundity and population size in Southern New England (Goldstein, Zarrella-Smith, and Pugh 2022) while the fishery in the central Gulf of Maine has increased dramatically (Le Bris et al. 2018). Temperatures have increased in the lobster's inshore range, and evidence suggests that the productivity of the stock is severely compromised with increased thermal stress and disease (Atlantic States Marine Fisheries Commission (ASMFC) 2025). The assemblage of predators of American lobster has shifted with warming conditions in the Gulf of Maine, with benthic and demersal predators, such as black sea bass (*Centropristis striata*) and spiny dogfish (*Squalus acanthias*) becoming increasingly present throughout the lobster's range, taking the place of traditional predators, such as Atlantic cod (*Gadus morhua*). The effects of invasive and range-expanding species may also lead to greater predation risk for juvenile lobsters (Goldstein et al. 2025).

Life History Synopsis

American lobsters inhabit the coastal and oceanic waters of the Atlantic Ocean from Labrador to North Carolina. They are widely distributed within most of the coastal waters in the Northeast U.S. with some major offshore concentrations along the outer edge of the Continental Shelf and upper slope between Georges Bank and Delaware Bay (Mackenzie and Moring 1985). Spawning takes place during a 4 month period from late May through much of September, when females leave their solitary shelters about 7 days before molting to share the shelter of a dominant male (Factor 1995) (Mackenzie and Moring 1985). Lobster larvae are planktonic for 6-8 weeks (Factor 1995) and travel from surface waters offshore to inshore areas of Southern New England and the Gulf of Maine. Larval distribution is affected by the distribution of spawning females and by surface current velocity and direction, temperature, salinity, light intensity, and hydrostatic pressure (Katz, Cobb, and Spaulding 1994) (Harding and Trites 1988) (Hudson and Fradette 1993; Mackenzie and Moring 1985). Larval survival is likely affected by pollution of the marine

environment by human activity, especially coastal waters (Connor 1972). There is an inverse relationship between temperature and the time from hatching to the postlarval stage (when they settle). Larvae take as few as 11 days at 22°C and as many as 54 days at 10°C to reach the postlarval stage. It is speculated that thermoclines are a negative cue for larvae/post larvae settlement (Factor 1995; Templeman 1936). Juvenile lobsters utilize shelters (rocks and vegetation) in the benthic habitat, and may also burrow in mud (Wahle and Steneck 1991). Juveniles are omnivorous but also scavenge for food. Much of their diet consists of bottom invertebrates, crabs, urchins, mussels, polychaetes, periwinkles, and sea stars, but they may also consume fish and plants (Lawton 1987). Adult lobsters are also considered omnivores and scavengers . However, their diet is dominated by mollusks (especially mussels), echinoderms, and polychaetes (Elner and Campbell 1987). Juveniles and adults prefer bottom water temperatures of 8-14°C (Mackenzie and Moring 1985). Inshore, adult lobsters appear to have a limited home range of about 2.2 km, however offshore lobsters tend to migrate greater distances (up to 345 km in 71 days) (Mackenzie and Moring 1985). American lobster are managed by the Atlantic States Marine Fisheries Commission as three stock units based on life history differences: Gulf of Maine, Georges Bank, and southern New England (Atlantic States Marine Fisheries Commission (ASMFC) 2025).

Supplemental Scores

Exposure Timeseries

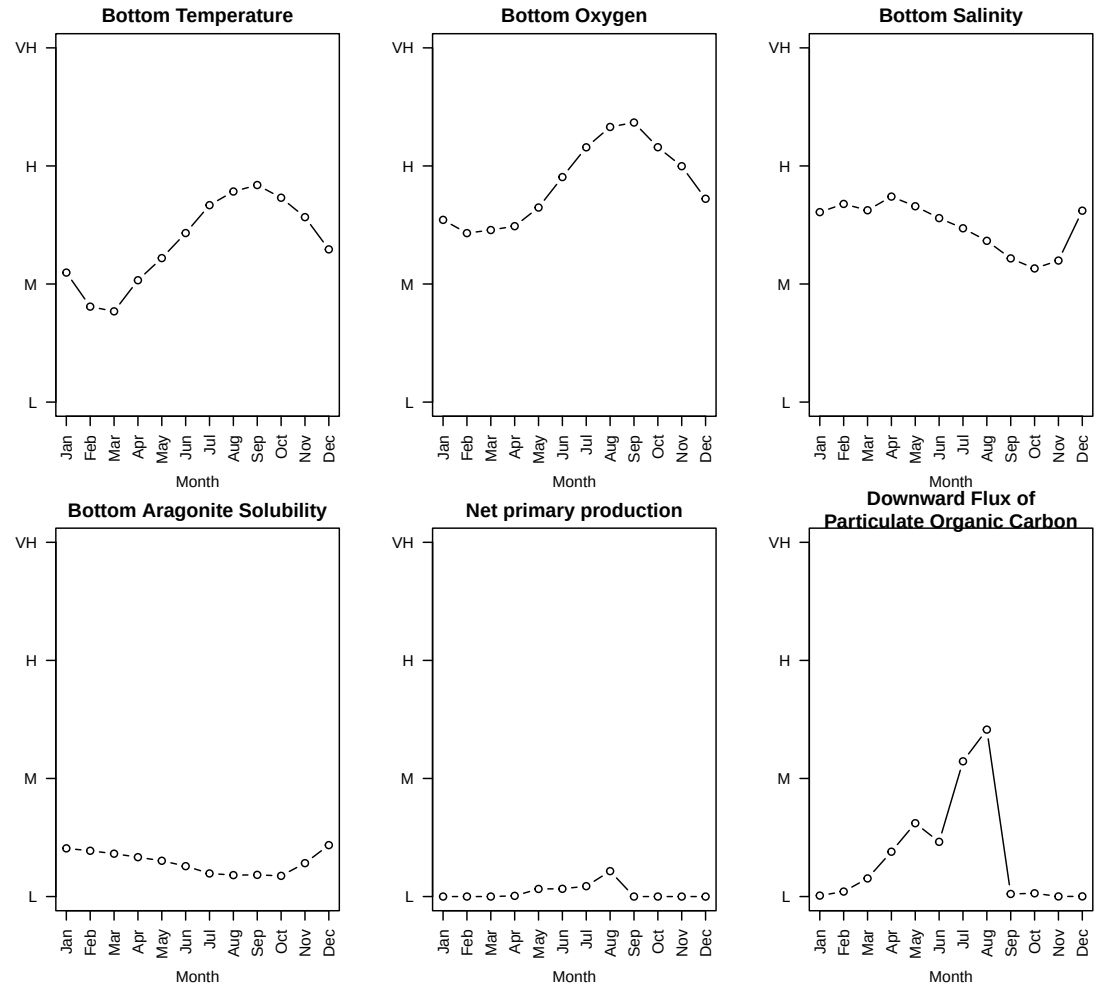


Figure 7: Monthly Timeseries of each Exposure Factor for American Lobster. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure scores ranged from Moderate to Low (Figure 8). Scores were Low from February to April, corresponding to a decline in bottom temperature and oxygen exposure (Figure 8, Figure 7). Scores were Moderate for the remainder of the year (Figure 8).

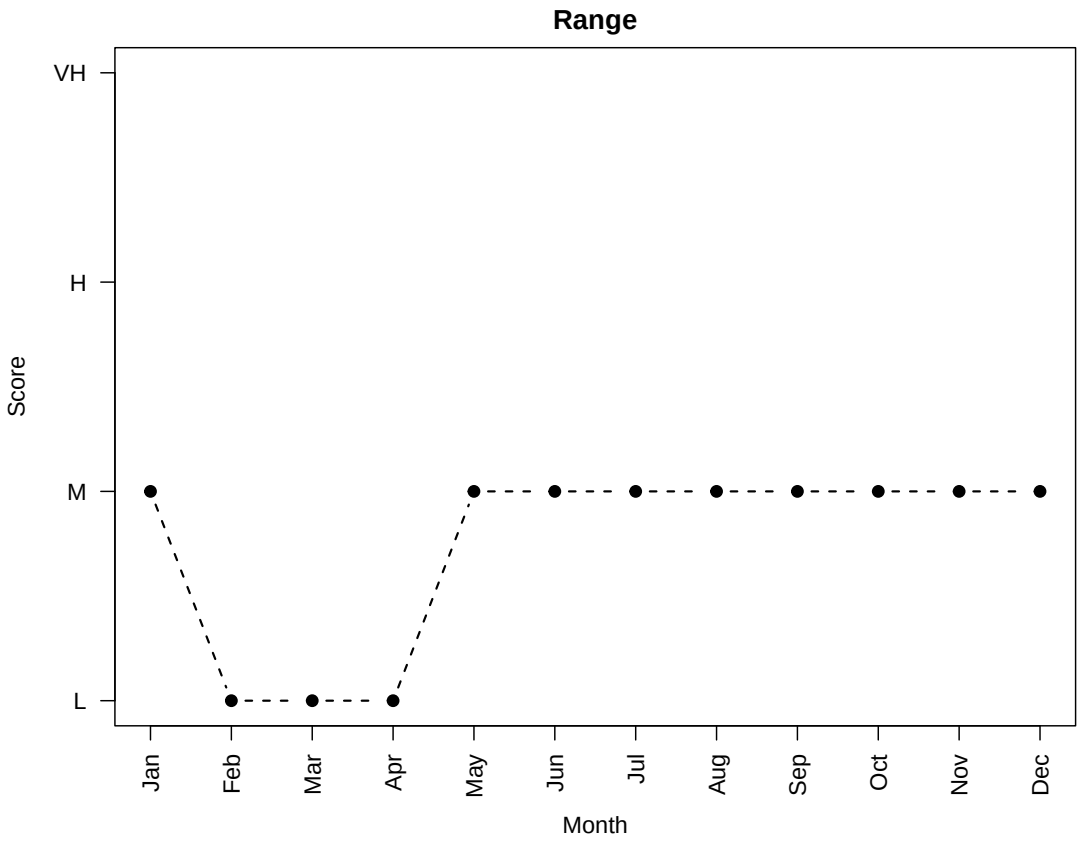


Figure 8: Timeseries of Total Exposure for American Lobster

Exposure due to Significant Factors

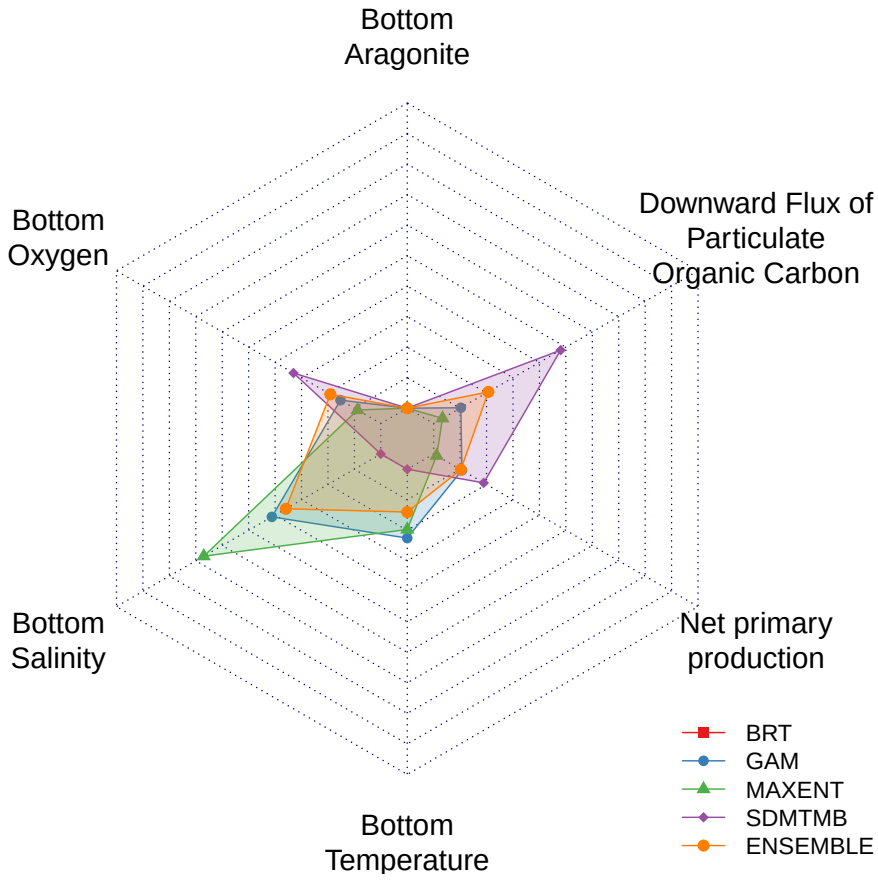


Figure 9: Dynamic Factors driving American Lobster distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, and salinity, in addition to net primary production and downward carbon flux were all important dynamic drivers of lobster distributions (Figure 9). Lobsters are omnivorous scavengers (Elner and Campbell 1987). They prefer colder waters and utilize rocky and vegetative habitats for cover (Wahle and Steneck 1991). When only these factors were used to calculate exposure, scores remained Moderate (1.99 +/- 0.96) (Figure 11, Figure 10).

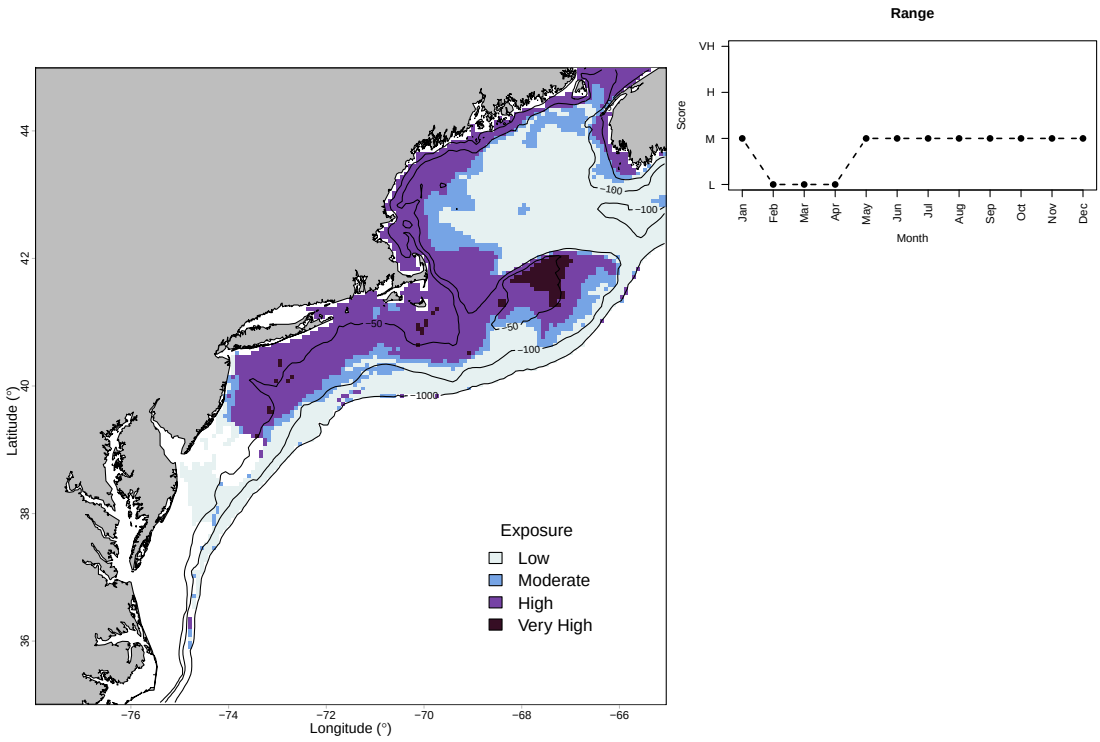


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to American Lobster distributions

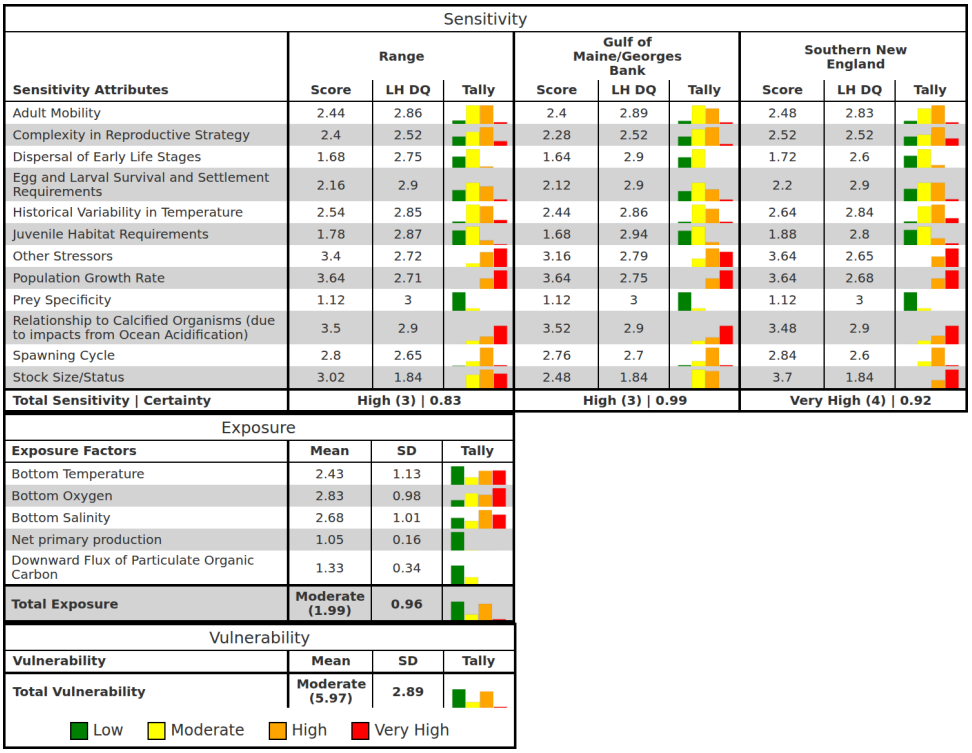


Figure 11: As in Figure 1 for American Lobster, but with only significant Exposure factors and the resulting Vulnerability Scores.

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